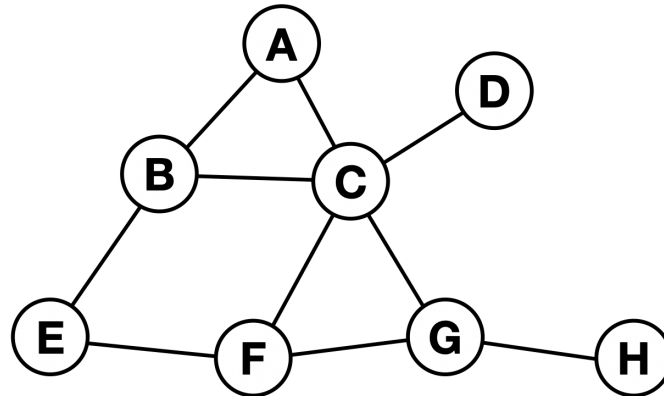


Multimodal Deep Learning

Exercise Sheet 5

Date: 17th of June, 2025

Exercise 1: Graph representations



Given is above's graph with the vertices $\mathcal{V} = \{\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}, \mathbf{E}, \mathbf{F}, \mathbf{G}, \mathbf{H}\}$.

- (a) Write down its representation as list of edges $\mathcal{E}$.
- (b) Write down its representation as the adjacency matrix $\mathcal{A}$.

Exercise 2: Graph scaling

Assume a social graph of 1,000,000 members (nodes) and that each node is connected to 200 randomly selected other nodes.

- (a) How large is the adjacency matrix of this graph?
- (b) How many entries of the adjacency matrix are non-zero?
- (c) Given a random node, estimate how many hops are required until 50% of the network can be reached?
- (d) Why may this estimation not hold in real-world social graphs?
- (e) How would above's scaling behavior affect the application of graph convolutional neural networks?

Exercise 3: Homogeneous vs. heterogeneous graph convolutional networks

In the lecture we introduced the message passing update rule for a given node of graph convolutional neural networks as:

$$v_i^{(l+1)} = \sigma \left(\sum_{j \in \mathcal{N}_i} \frac{1}{c_i} W^{(l)} v_j^{(l)} + W_0^{(l)} v_i^{(l)} \right) \quad (1)$$

However, for homogeneous graphs it is often written as:

$$v_i^{(l+1)} = \sigma \left(W^{(l)} \sum_{j \in \mathcal{N}_i} \frac{1}{c_i} v_j^{(l)} + W_0^{(l)} v_i^{(l)} \right) \quad (2)$$

Finally, recall that the node update rule for heterogeneous graph convolutional neural networks is:

$$v_i^{(l+1)} = \sigma \left(\sum_{r \in R} \sum_{j \in \mathcal{N}_i^r} \frac{1}{c_{i,r}} W_r^{(l)} v_j^{(l)} + W_0^{(l)} v_i^{(l)} \right) \quad (3)$$

- What do the variables $v_i^{(l)}, v_j^{(l)}, \mathcal{N}_i, W^{(l)}, W_0^{(l)}, \sigma$ and $\sum_{j \in \mathcal{N}_i}$ correspond to?
- How do the networks described by Equations 1 and 2 differ? Do these differences have a large impact on their performance?
- What do the variables $W_r^{(l)}$ and $\mathcal{N}_i^r$ correspond to?
- Why is Equation 1 used as basis when extending a homogeneous to a heterogeneous graph neural network instead of Equation 2?

Exercise 4: GNNs, CNNs and transformers

The lecture discussed that CNNs and transformers are both special cases of graph convolutional neural networks.

- What is the adjacency matrix for a given pixel and a 3×3 convolutional kernel?
- Does a CNN rather resembles a homogeneous or a heterogeneous graph convolutional neural network (cf. Equations 2 and 3 above)?
- How would the connectivity graph of a vanilla transformer look like?
- What network components are missing to extend a graph convolutional neural network as described in Equation 2 to a transformer architecture?